

Travelling Salesperson Problem (TSP) - Heuristic Search (Nearest Neighbor)

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# Cost matrix between cities
cost = [[0,10,15,20],
        [10,0,35,25],
        [15,35,0,30],
        [20,25,30,0]]

n = len(cost)
visited = [0]      # Start city
total = 0          # Total cost

while len(visited) < n:
    city = visited[-1]

    nxt = min(
        (cost[city][i], i)
        for i in range(n)
        if i not in visited
    )

    total += nxt[0]
    visited.append(nxt[1])

# Return to starting city
total += cost[visited[-1]][0]
visited.append(0)

print("Path:", visited)
print("Minimum Cost:", total)
Output: Path=[0,1,3,2,0], Minimum Cost=80
```